Kartik Padmanabhan

+91-97787-13724 | kartikp9595@gmail.com | linkedin.com/in/kartik-padmanabhan | github.com/Kartik-95

Summary

Data Science undergraduate with hands-on experience in Python, SQL, and machine learning, delivering academic projects in churn prediction, explainable AI, and real-time computer vision. Seeking a Data Analyst role to turn data into clear, actionable business insights.

Education

Mar Athanasius College of Engineering 2023 – Present
B.Tech in Computer Science and Engineering (Data Science) — CGPA: 8.05 *Kothamangalam, Kerala*

Govt Higher Secondary School 2021 – 2023
Biology Science — Percentage: 99.25% *Vayakkara, Kerala*

Projects

Customer Churn Prediction & Explainable AI (XAI) App | *Python, Pandas, Scikit-learn, XGBoost, SHAP, Streamlit* 2026

- Developed and deployed an Explainable AI web app in Streamlit to predict telecom customer churn on ∼7,000 records with 26% class imbalance.
- Tuned an XGBoost model via hyperparameter search and class-weighting, improving ROC-AUC from 0.897 to 0.912 and reaching 0.787 PR-AUC under 5-fold cross-validation.
- Integrated SHAP explainability with per-prediction and global visualizations to help stakeholders interpret key churn drivers.

Intelligent Surveillance Framework for Crowd Detection and Alerting | *Python, YOLOv8, CSRNet, Flask* 2026

- Engineered a hybrid crowd-analysis pipeline that routes sparse scenes to YOLOv8 for object detection and dense crowds to CSRNet for density estimation.
- Achieved a Mean Absolute Error (MAE) of 10.6 on standard crowd-counting benchmarks, maintaining reliable counts across both sparse and dense environments.
- Shipped a Flask dashboard delivering real-time crowd counts and threshold-based alerts for monitoring staff.

Technical Skills

Languages: Python, SQL, HTML, CSS
Libraries: Pandas, NumPy, Scikit-learn, Matplotlib
Tools & Development Environments: Git, MySQL, VS Code, Jupyter Notebook, Google Colab, Excel, Power BI, Flask
Machine Learning & Data Science: Machine Learning, Data Analytics, Logistic Regression, XGBoost
Domains: Data Analysis, Machine Learning, Data Visualization

Achievements

Ranked in the Top 12 of 60 teams — .hack() 2025 Hackathon.

Certifications

Tata GenAI Powered Data Analytics Job Simulation — Forage (Jan 2026)

Deloitte Data Analytics Job Simulation — Forage (Mar 2026)

Languages

English, Malayalam, Hindi, Tamil